

In RAG we trust?

Alessandro Sarchi

Department of Computer Science, University of Milan, Via Celoria 18,
Milan, 20133, Italy.

Contributing authors: alessandro.sarchi02@gmail.com;

Abstract

Retrieval-augmented generation (RAG) is vulnerable to corpus poisoning: an adversary who plants contradictory passages into the retrieved context can mislead a language model without modifying its weights. This paper evaluates two inference-time defences, chain-of-thought (CoT) and vigilant prompting, against a binary full-poisoning attack ($\mathbf{r} \in \{\mathbf{0}, \mathbf{1}\}$) on two benchmarks: FEVER (claim verification) and HotpotQA (multi-hop question answering). Two open-weight models (*Qwen3.5-2B* and *Gemma-4-E2B-IT*) are evaluated with $N = \mathbf{75}$ examples per condition, $k = \mathbf{10}$ retrieved passages, and greedy decoding ($\tau = \mathbf{0}$). Poisoning degrades token F1 in every HotpotQA condition (drops of 0.04–0.09) and degrades both models on FEVER under all prompt types (accuracy drops of 0.04–0.11). Qwen3.5’s smaller accuracy degradation under standard and vigilant prompting is explained by a persistently high hallucination rate ($\geq \mathbf{0.58}$ in clean conditions), indicating label-prior reliance rather than evidence-grounded inference; when CoT forces context engagement, poisoning causes the largest observed degradation ($\Delta F1 = -0.17$). Neither CoT nor vigilant prompting provides consistent protection across both model families and tasks, suggesting that inference-time prompt defences alone are insufficient to secure small open-weight RAG pipelines against targeted corpus poisoning.

1 Introduction

Retrieval-augmented generation (RAG) [1] has become a standard paradigm for knowledge-intensive NLP tasks, reducing parametric hallucination by conditioning generation on externally retrieved documents. This architecture is widely adopted in fact-verification [2] and question-answering pipelines, where the quality of retrieved evidence directly shapes the model’s output.

This reliance on an external corpus introduces a vulnerability: an adversary who inserts misinformation into the retrieval index can mislead the model without modifying its weights [3, 4]. Two inference-time defences have been proposed: chain-of-thought prompting [5], which encourages step-by-step evidence evaluation, and vigilant prompting [6], which instructs the model to flag inconsistencies in the retrieved context. Whether either strategy provides meaningful protection across model families and task types remains an open empirical question.

This paper addresses that question under a strong adversarial assumption: the attacker has *already bypassed the retriever* and poisoned passages are already present in the model context. Two open-weight instruction-tuned models ($\approx 2B$ parameters) are evaluated on FEVER [7] and HotpotQA [8] under a binary poisoning condition ($r \in \{0, 1\}$) and three prompt types (standard, chain-of-thought, vigilant), holding the retriever fixed to Yuan-embedding-2.0-en [9]. Poisoning is performed via LLM negation, in which each gold evidence passage is rewritten into a fluent direct contradiction.

2 Research Question and Methodology

The evaluation pipeline is organized into three sequential phases: *retrieval*, *generation*, and *evaluation*.

2.1 Retrieval

2.1.1 Corpus construction

For each example, a distractor pool is drawn from the union of all gold passages associated with every other example in the dataset (*global gold pool*). Unlike a fixed-size random sample of distractors, the global pool is both large and factually grounded: its passages are gold evidence from real examples, making the similarity-based retrieval step non-trivial without introducing random noise. Embeddings for the full pool are computed once and reused across examples.

The model context is composed by concatenating $k_{\text{dist}} = \max(0, k - n_{\text{gold}})$ passages retrieved from the global pool and n_{gold} evidence passages (original at $r = 0$, LLM-negated at $r = 1$), giving exactly k passages in total. This *injection* design holds retrieval quality constant across conditions, so observed differences in accuracy are attributable solely to the content of the injected evidence.

2.1.2 Poisoning

Gold evidence passages are replaced with fluent contradictions generated by a language model (*LLM negation*). The poison condition is binary: at $r = 0$ all evidence passages are original; at $r = 1$ all evidence passages are replaced by LLM-generated negations. For FEVER, every gold passage associated with a claim is negated at $r = 1$. For HotpotQA, exactly one supporting fact per example is negated: a single broken link in the multi-hop reasoning chain is sufficient to render the correct answer unrecoverable.

Negation is performed by prompting a fixed instruction-tuned model (*Qwen3.5-2B*) to rewrite a passage so that its meaning is directly reversed, producing a single

fluent sentence about the same subject. Negations are pre-computed once and stored on disk before any evaluation model is run, ensuring that all evaluated models face exactly the same poisoned passages and enabling fair cross-model comparison [3].

2.1.3 Embedding and caching

All passages and example strings are encoded with *Yuan-embedding-2.0-en* [9], the top-ranked model on the MTEB Retrieval leaderboard at the time of the experiments [10]. Each text is mapped to a dense vector of dimension $d = 1024$; vectors are L2-normalised prior to indexing and cached on disk.

2.1.4 Top- k retrieval

The global pool embeddings are indexed in a FAISS `IndexFlatIP` structure [1] for exact inner-product search; exact search is preferable over the approximate HNSW used in [1] given the manageable pool size. The k_{dist} most similar distractor passages are retrieved and combined with the n_{gold} injected passages to form the $k = 10$ passages presented to the generation phase.

2.2 Generation

2.2.1 Language models

Two open-weight instruction-tuned decoder-only models are compared: *Qwen3.5-2B* and *Gemma-4-E2B-IT*, both in the 2B-parameter class. Both are loadable locally via Hugging Face, keeping inference feasible on a single consumer GPU or CPU. They span different providers (Alibaba and Google), following [4] to ensure findings are not artefacts of a single model family. All models run with temperature $\tau = 0$ (greedy decoding); responses are cached on disk so repeated runs incur no inference cost.

2.2.2 Prompt design

Three prompt variants are evaluated for each task. For FEVER claim verification:

- **Standard:** the model classifies the claim directly from the numbered retrieved passages, with no additional reasoning instructions.
- **Chain-of-thought (CoT):** the model is instructed to reason step by step through each passage before emitting a **Final Label**, following the prompting strategy of [5]. Although [5] report CoT gains primarily on models above 100B parameters, we include it as an empirical test of whether structured reasoning provides any benefit in the small-model regime.
- **Vigilant:** the model is first asked whether the retrieved passages are mutually consistent; if a contradiction is detected, it is noted before the final label is produced. This template operationalises the vigilant prompting defence proposed by [6].

Analogous templates (*standard_qa*, *cot_qa*, *vigilant_qa*) are used for HotpotQA, replacing the claim with the question and the three-way label with a short free-form answer.

2.2.3 Output parsing

For FEVER, label extraction proceeds in two stages: (i) a regex scan for a structured **Final Label:** marker (produced by CoT and vigilant prompts), and (ii) a keyword fallback in order of specificity: NOT ENOUGH INFO, then SUPPORTS, then REFUTES. Responses matching no pattern default to NOT ENOUGH INFO. For HotpotQA, the extractor scans for a **Final Answer:** / **Answer:** marker (including bold-markdown variants); if absent, it falls back to the inline pattern *the answer is ...* to recover CoT-style responses, then to the stripped full response. For the vigilant template, a secondary parser checks the consistency section for explicit contradiction language (*contradict, conflict, inconsistent, disagree*).

2.3 Evaluation

2.3.1 FEVER metrics

Three metrics are reported for every experimental condition on FEVER:

- **Accuracy:** fraction of predicted labels matching the gold label.
- **Macro-F1:** macro-averaged F1 over the three FEVER labels, compensating for the dominance of the SUPPORTS class.
- **Hallucination rate:** fraction of NOT ENOUGH INFO examples for which the model nonetheless predicts SUPPORTS or REFUTES, asserting a verdict not grounded in the retrieved context.

2.3.2 HotpotQA metrics

Two standard metrics are adopted from Yang et al. [8], who follow Rajpurkar et al. [11]:

- **Exact match:** 1.0 if the normalised prediction equals the normalised gold answer, 0.0 otherwise.
- **Token F1:** token-level F1 between the normalised prediction and the gold answer, after stripping punctuation, articles, and extra whitespace.

A third metric is introduced in this work:

- **QA hallucination rate:** fraction of predicted answers for which fewer than 50% of their tokens appear in the retrieved passages (token precision < 0.5), after applying the same normalisation as exact match and token F1. Operationalises grounding failure; precision is preferred over F1 to avoid penalising short but fully grounded answers against long passages.

2.3.3 Experimental variables

The following factors are varied across the full experimental design:

- **Poisoning condition:** $r \in \{0, 1\}$ (clean vs fully poisoned).
- **Prompt type:** standard, chain-of-thought, vigilant.
- **Model:** Qwen3.5-2B, Gemma-4-E2B-IT.

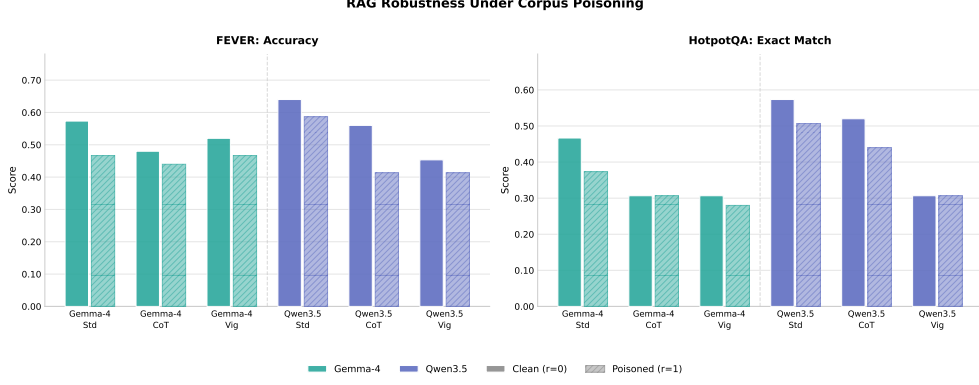


Fig. 1 Accuracy (FEVER, left) and exact match (HotpotQA, right) under clean ($r = 0$, solid) and poisoned ($r = 1$, hatched) conditions across all model and prompt-type combinations. The dashed vertical line separates model groups.

- **Dataset:** FEVER (claim verification) vs HotpotQA (multi-hop QA).

All conditions share a fixed random seed ($s = 42$) and the same retriever, yielding $2 \times 3 \times 2 = 12$ conditions per dataset (24 total across both datasets).

3 Experimental Results

3.1 Experiment 1: Main Poisoning Experiment (FEVER)

Figure 1 and Table 1 summarise results for $N = 75$ FEVER examples ($2 \times 3 \times 2 = 12$ conditions, $\tau = 0$).

Gemma-4 degrades most under standard prompting ($\Delta\text{acc} = -0.11$, $\Delta\text{F1} = -0.11$); CoT and vigilant prompting provide equal accuracy protection (-0.04 each), with vigilant showing the smaller macro-F1 drop (-0.07 vs -0.10). Qwen3.5’s accuracy drops are smaller under standard (-0.05) and vigilant (-0.04) prompting, but its hallucination rate is persistently high (Table 1), indicating label-prior reliance: the model disproportionately predicts SUPPORTS or REFUTES on NOT ENOUGH INFO examples regardless of poisoning condition.

The most damaging condition is CoT prompting for Qwen3.5, which produces the largest macro-F1 drop (-0.169): when instructed to reason step-by-step, the model engages with the contradictory evidence rather than defaulting to label priors.

Counter-intuitively, Gemma-4’s hallucination rate *decreases* under poisoning in CoT ($0.31 \rightarrow 0.08$) and vigilant ($0.15 \rightarrow 0.08$) conditions, as assertive poisoned passages push the model away from NOT ENOUGH INFO toward a definite but incorrect verdict. Vigilant prompting detects contradictions in only $\approx 19\%$ of Gemma-4 contexts and 0% of Qwen3.5 contexts, explaining its limited accuracy benefit.

Table 1 FEVER secondary metrics: macro-F1 and hallucination rate.

Model	Prompt	Macro-F1			Hallucination rate		
		$r=0$	$r=1$	Δ	$r=0$	$r=1$	Δ
Gemma-4	Standard	0.539	0.432	-0.107	0.808	0.846	+0.038
	CoT	0.475	0.373	-0.102	0.308	0.077	-0.231
	Vigilant	0.483	0.415	-0.068	0.154	0.077	-0.077
Qwen3.5	Standard	0.626	0.584	-0.042	0.577	0.423	-0.154
	CoT	0.533	0.364	-0.169	0.115	0.231	+0.115
	Vigilant	0.413	0.400	-0.013	0.885	0.769	-0.115

Table 2 HotpotQA secondary metrics: token F1 and QA hallucination rate.

Model	Prompt	Token F1			QA hallucination rate		
		$r=0$	$r=1$	Δ	$r=0$	$r=1$	Δ
Gemma-4	Standard	0.580	0.536	-0.044	0.053	0.093	+0.040
	CoT	0.465	0.423	-0.042	0.053	0.080	+0.027
	Vigilant	0.511	0.421	-0.090	0.147	0.213	+0.067
Qwen3.5	Standard	0.684	0.606	-0.079	0.133	0.133	0.000
	CoT	0.655	0.568	-0.088	0.147	0.133	-0.013
	Vigilant	0.457	0.403	-0.054	0.187	0.160	-0.027

3.2 Experiment 2: HotpotQA Cross-Dataset Evaluation

To test whether FEVER findings generalise to multi-hop open-domain QA, the main poisoning experiment is replicated on HotpotQA. The design mirrors Experiment 1: $N = 75$ examples, $k = 10$, both models, three QA prompt variants (standard_qa, cot_qa, vigilant_qa), and both poisoning conditions ($2 \times 3 \times 2 = 12$ conditions, $\tau = 0$). Metrics follow the SQuAD evaluation protocol [11]: exact match, token F1, and QA hallucination rate (Table 2).

Token F1 decreases in every condition (Figure 1, Table 2), confirming that poisoning consistently degrades answer quality. Exact-match degradation is more varied: two conditions (Gemma-4 CoT, Qwen3.5 Vigilant) show zero EM drop, though their token F1 still declines (-0.042 and -0.054 respectively), indicating partial rather than complete robustness. Gemma-4 Vigilant suffers the largest token-F1 drop (-0.090), and no single prompt strategy is consistently best across both models and metrics.

The contrast with FEVER is informative: on FEVER, near-zero accuracy drops are possible when the model relies on label priors; on HotpotQA no such shortcut exists, because the task requires a free-form answer grounded in multi-hop evidence. The consistent degradation therefore provides stronger evidence that corpus poisoning genuinely disrupts evidence integration.

4 Concluding Remarks

Summary of findings.

Corpus poisoning degrades token F1 in every HotpotQA condition (up to -0.09) and accuracy in every FEVER condition (up to -0.11). Qwen3.5’s smaller accuracy drops under standard and vigilant prompting are explained by persistently high hallucination rates (≥ 0.58 in clean conditions), indicating label-prior reliance; when CoT forces context engagement, poisoning causes the largest degradation observed ($\Delta F1 = -0.169$). Neither chain-of-thought nor vigilant prompting provides consistent protection across both model families and tasks: CoT reduces Gemma-4’s FEVER accuracy drop but amplifies Qwen3.5’s macro-F1 loss, and vigilant prompting detects contradictions in only $\approx 19\%$ of Gemma-4 contexts and 0% of Qwen3.5 contexts. These results indicate that inference-time prompt defences are insufficient to robustify small open-weight models against targeted corpus poisoning.

Limitations and future work.

The primary limitation of this study is sample size: $N \leq 60$ examples per condition limits statistical power and makes it difficult to distinguish genuine model differences from sampling variance. Future work should replicate the experiment on larger samples ($N \geq 500$) and extend the model range to instruction-tuned models above 7B parameters, where chain-of-thought gains are more reliably observed [5]. A second open direction is improving contradiction detection: if the vigilant prompt consistently fails to flag inconsistencies, a more explicit grounding mechanism, such as passage-level entailment scoring or evidence-backed verification [2], may be needed before inference-time defences can be considered effective. Third, self-consistency decoding [12] was not evaluated in this work: applying it would require temperature $\tau > 0$ and sampling multiple responses per example, multiplying inference costs by a factor of 5–40 per condition, which was computationally prohibitive under the hardware constraints of this project. Finally, the interaction between retriever quality and generation robustness warrants dedicated study: the injection design of this work deliberately holds retrieval quality constant, whereas real deployments face joint degradation of both components.

Use of AI Tools

Claude Sonnet 4.6 (Anthropic) was used as an AI assistant in four roles: *test generation* (unit tests for the TDD workflow, reviewed by the author); *design critique* (surfacing potential weaknesses in design proposals, with all decisions made by the author); *literature navigation* (locating relevant sections in cited papers); and *prose review* (sentence-level fluency feedback). The research question, experimental design, implementation, data analysis, and interpretation of results were carried out exclusively by the author.

References

- [1] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler,

- H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474 (2020)
- [2] Singal, R., Patwa, P., Patwa, P., Chadha, A., Das, A.: Evidence-backed fact checking using RAG and few-shot in-context learning with LLMs. In: *Proceedings of the Seventh Fact Extraction and VERification Workshop (FEVER)*, pp. 91–98 (2024)
- [3] Zhong, Z., Huang, Z., Wettig, A., Chen, D.: Poisoning retrieval corpora by injecting adversarial passages. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 13764–13775. Association for Computational Linguistics, Singapore (2023)
- [4] Zhou, Y., Liu, Y., Li, X., Jin, J., Qian, H., Liu, Z., Li, C., Dou, Z., Ho, T.-Y., Yu, P.S.: Trustworthiness in Retrieval-Augmented Generation Systems: A Survey (2024)
- [5] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E.H., Le, Q.V., Zhou, D.: Chain-of-thought prompting elicits reasoning in large language models. In: *Advances in Neural Information Processing Systems*, vol. 35, pp. 24824–24837 (2022)
- [6] Pan, Y., Pan, L., Chen, W., Nakov, P., Kan, M.-Y., Wang, W.Y.: On the risk of misinformation pollution with large language models. In: *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 1389–1403. Association for Computational Linguistics, Singapore (2023)
- [7] Thorne, J., Vlachos, A., Christodoulopoulos, C., Mittal, A.: FEVER: a large-scale dataset for fact extraction and VERification. In: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 809–819. Association for Computational Linguistics, New Orleans, Louisiana (2018)
- [8] Yang, Z., Qi, P., Zhang, S., Bengio, Y., Cohen, W.W., Salakhutdinov, R., Manning, C.D.: HotpotQA: A dataset for diverse, explainable multi-hop question answering. In: *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 2369–2380. Association for Computational Linguistics, Brussels, Belgium (2018)
- [9] IEITYuan: Yuan-embedding-2.0-en. Hugging Face. Accessed: 2025-05-17 (2024). <https://huggingface.co/IEITYuan/Yuan-embedding-2.0-en>
- [10] Muennighoff, N., Tazi, N., Magne, L., Reimers, N.: MTEB: Massive Text Embedding Benchmark (2023)
- [11] Rajpurkar, P., Zhang, J., Lopyrev, K., Liang, P.: SQuAD: 100,000+ questions

for machine comprehension of text. In: Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing, pp. 2383–2392. Association for Computational Linguistics, Austin, Texas (2016)

- [12] Wang, X., Wei, J., Schuurmans, D., Le, Q.V., Chi, E.H., Narang, S., Chowdhery, A., Zhou, D.: Self-consistency improves chain of thought reasoning in language models. In: International Conference on Learning Representations (2023)